

E4585560_1

The equation $x^4 + x^2 + x + y^3 + y + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$x^4 + x^2 + x + y^3 + y + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 - 5V_k^2$ on the curve (1), and the fact that an odd prime p with $(5/p) = -1$ divides $U_k^2 - 5V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm form $a^2 - 5b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(5/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 - 5b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv 5(bc)^2 \equiv 5 \pmod{p}$, so 5 is a square modulo p . As $(5/p) \neq 0$, it is a non-zero square, so $(5/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 - 5b^2 \neq 0$, and $v_p(a^2 - 5b^2)$ is even for every odd prime p with $(5/p) = -1$.*

Proof. Since 5 is not the square of an integer, $a^2 - 5b^2 \neq 0$. Let p be an odd prime with $(5/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 - 5b'^2$ by Lemma 2. Since $a^2 - 5b^2 = p^{2k}(a'^2 - 5b'^2)$, we get $v_p(a^2 - 5b^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{20}$, or with $n' \equiv -n \pmod{20}$. Then $(5/n') = (5/n)$.*

Proof. Write $5 = \varepsilon 2^e q$ with $\varepsilon = 1$, $e = 0$ and $q = 5$. By multiplicativity and the reciprocity laws,

$$\left(\frac{5}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since 20 is divisible by $4q$, the first claim follows. If $n' \equiv -n \pmod{20}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. This proves the second claim. $\square$

Lemma 5. *Let n be a non-zero integer, and let s , u and m be positive integers such that every prime factor of s divides 10, both $2s$ and $20s$ divide m , u is odd, $(5/u) = -1$, and $n \equiv su \pmod{m}$. Then there is an odd prime p with $(5/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by 20, n' is odd and $n' \equiv u \pmod{20}$. Since $|n'| \equiv \pm u \pmod{20}$, Lemma 4 gives $(5/|n'|) = (5/u) = -1$. By multiplicativity, the product of $(5/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(5/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 10$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x^2 + x + y^3 + y + 3$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= x^2 + 2x + 4y + 4, \\ T_1 &= x^4 + 4x^3 - 4x^2y + 12x^2 - 8xy + 16x + 16y^2 - 16y + 32, \\ U_1 &= x^3 + 3x^2 - 22x + 44, \\ V_1 &= x^2 - 10x + 20, \\ G_1 &= 64, \\ A_1 &= -11x + 83, \\ B_1 &= 11x^2 + 60x - 111, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= x - y + 1, \\ T_2 &= 4x^2 + 4xy + 8x + 4y^2 + 4y + 8, \\ U_2 &= 2x^2 + x + 5, \\ V_2 &= x - 1, \\ G_2 &= -4, \\ A_2 &= 1, \\ B_2 &= -2x - 3. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1 T_1 - U_1^2 + 5V_1^2 = G_1 F, \tag{2}$$

$$H_2 T_2 - U_2^2 + 5V_2^2 = G_2 F, \tag{3}$$

$$A_1 U_1 + B_1 V_1 = 2^3 \cdot 179, \tag{4}$$

$$A_2 U_2 + B_2 V_2 = 2^3. \tag{5}$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_k T_k = U_k^2 - 5V_k^2 \quad (k = 1, 2), \quad (6)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 1600^2 residue pairs modulo 1600, we find that (1) has exactly 1280 solutions modulo 1600, and that at each of them $H_1 \bmod 1600$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 1600$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(5/u) = -1$; here for $s = 1$, $2s = 2$ and $20s = 20$ divide 1600, and for $s = 5$, every prime factor of s divides 10, and $2s = 10$ and $20s = 100$ divide 1600. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(5/u) = -1$; here every prime factor of each $s \in \{5, 20, 80\}$ divides 10, and $2s$ and $20s$ divide 1600 for each such s .

Step 2: signs. *The value of H_1 .* By (4), U_1 and V_1 are not both zero, so $U_1^2 - 5V_1^2 \neq 0$ by Lemma 3, and (6) gives $H_1 \neq 0$.

The value of H_2 . By (5), U_2 and V_2 are not both zero, so $U_2^2 - 5V_2^2 \neq 0$ by Lemma 3, and (6) gives $H_2 \neq 0$.

Step 3: primes p with $(5/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(5/p) = -1$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (6) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 - 5V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 - 5V_k^2$ by (6), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 2^3 \cdot 179$. But p is odd, and $p \neq 179$ because $(5/179) = 1$, a contradiction. For $k = 2$, we have $R_2 = 2^3$. But p is odd, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{1600}$ with s and u as in Lemma 5, and $H_k \neq 0$ by Step 2. Hence Lemma 5 applies to $n = H_k$, so there is an odd prime p with $(5/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 1600 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
3	267	527	827	1107	1387
15 = 5 · 3	283	547	843	1115 = 5 · 223	1403
27	307	563	847	1123	1407
43	315 = 5 · 63	587	867	1147	1423
63	323	603	883	1163	1427
67	347	627	907	1167	1443
83	363	643	915 = 5 · 183	1187	1467
107	383	667	923	1203	1483
115 = 5 · 23	387	683	947	1215 = 5 · 243	1487
123	403	703	963	1227	1507
127	427	707	987	1243	1515 = 5 · 303
143	443	715 = 5 · 143	1003	1267	1523
147	447	723	1023	1283	1547
163	463	747	1027	1307	1563
187	467	763	1043	1315 = 5 · 263	1587
203	483	767	1067	1323	
207	507	783	1083	1343	
227	515 = 5 · 103	787	1087	1347	
243	523	803	1103	1363	
$\mathcal{R}_2$, the residues of H_2					
35 = 5 · 7	285 = 5 · 57	560 = 80 · 7	835 = 5 · 167	1085 = 5 · 217	1360 = 80 · 17
85 = 5 · 17	335 = 5 · 67	585 = 5 · 117	885 = 5 · 177	1135 = 5 · 227	1385 = 5 · 277
135 = 5 · 27	385 = 5 · 77	635 = 5 · 127	935 = 5 · 187	1185 = 5 · 237	1435 = 5 · 287
185 = 5 · 37	435 = 5 · 87	685 = 5 · 137	985 = 5 · 197	1235 = 5 · 247	1485 = 5 · 297
235 = 5 · 47	485 = 5 · 97	735 = 5 · 147	1035 = 5 · 207	1285 = 5 · 257	1535 = 5 · 307
260 = 20 · 13	535 = 5 · 107	785 = 5 · 157	1060 = 20 · 53	1335 = 5 · 267	1585 = 5 · 317